

UNIT 2

Stages of Growth and Growth Pattern

Fundamentals in Human Growth

Programme: MA Anthropology | **Course:** Fundamentals in Human Growth

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LEARNING OBJECTIVES

Once you have read this unit, you will:

- Be able to understand the concepts of growth and development
- Be acquainted with stages of growth
- Appreciate normal growth curve pattern, normal growth variation or variation from normal growth
- Get to know about ethnic and gender differences in growth curves

Section 2.0

INTRODUCTION

Primates generally have a slow reproductive turnover and drawn-out life histories. As part of this pattern, the pace of growth from birth to adulthood is retarded and in monkeys, apes and humans, at least, it is possible to recognize special features of the growth curve associated with the attainment of sexual maturity (puberty). The distinctive form of the human growth curve was first established in the eighteenth century. Sexual differences in the curve, originally denied, were confirmed in the 1880s, but the evolutionary significance of its shape only became clear after 1950 (Jaiswal, 2013).

The velocity, speed, or pace of development mirrors the child's state at a specific time more than does actual height, which depends largely on how much the child has grown throughout life. The blood and tissue concentrations of substances that change with age are more closely associated with the speed or velocity than the distance curve. In certain conditions, acceleration of growth rather than its velocity may reflect physiological events; the increase in secretion from the endocrine glands at adolescence, for example, is seen most clearly in an acceleration of growth.

The velocity of growth in height decreases from birth onwards, yet this decline is interrupted shortly before the end of the growth period. At this time, from 13 to 15 years of age in boys, there is a marked acceleration of growth—the **adolescent or pubertal growth spurt**. A slight increase in velocity, the **mid-growth spurt**, may also occur between about 6 and 8 years, providing a second peak on the general velocity curve. After maturity, the matured boy or girl becomes an important member of society.

Section 2.1

UNDERSTANDING GROWTH AND DEVELOPMENT

Understanding the growth and development of the human organism requires an understanding of nature and nurture, and the complex interplay between the two. The terms **growth** and **development** are actually two different terms having separate meanings, but most of the time they are used interchangeably. This creates a lot of confusion.

Growth	Development
Refers to quantitative changes —increase in size, length, weight, proportion.	Refers to qualitative changes —changes in kind, structure, or organization.
Can be observed and measured in the physical sense.	A complex process of integrating many structures and functions; may not be outwardly observable.
Enlargement of the body or its parts—heavier, larger, longer.	Integration and differentiation of structures and functions.

Thus, growth can be one of the components of the development process, or development in its quantitative aspect can be termed as growth.

Check Your Progress 1

1) Write short notes on understanding Growth and Development.

Section 2.2

STAGES OF GROWTH

The stages or phases of growth have been classified in different ways by different scholars. A simple classification of the postnatal period includes four stages: early childhood, later childhood, adolescence and puberty; or early childhood, middle childhood, late childhood and adolescence.

A detailed classification, proposed by **R.L. Scammon** (1930), is given below:

Table 2.1: Stages of Growth (Scammon, 1930)

Period	Stage	Sub-stage	Duration
Prenatal	Ovum		First 2 weeks
	Embryo		2 to 8 weeks
	Fetus		2 to 10 lunar months
Postnatal	Infancy	Neonatal	First 2 weeks
		Infancy	2 weeks to 1 year
	Childhood	Early	1 to 6 years
		Middle	6 to 9–10 years
		Late (Boys)	9–10 to 13–16 years
		Late (Girls)	9–10 to 12–15 years
	Puberty	Boys	Around 14 years
		Girls	Around 13 years
	Adolescence	Boys	14 to 20 years
		Girls	13 to 18–20 years
	Maturity		18–20 to 60 years
	Senility		After 60 years

Thus, the two major periods of growth are **prenatal** and **postnatal**. This is true not only for humans but also for all placental mammals.

Check Your Progress 2

2) Name the major periods of growth.

Section 2.3

PRENATAL GROWTH

Prenatal development includes all the changes that take place in the womb of the mother. Therefore it is also called "**intra-uterine development**" where the uterus is the internal environment for growth. It covers the period from fertilization to birth and comprises three distinct stages:

Stage	Period	Key Features
Zygote	Fertilization to 2 weeks	Resembles a homogeneous mass; about 0.1 mm in diameter
Embryo	2 weeks to 2 lunar months	Differentiation and regionalization; histogenesis; morphogenesis; childlike appearance by 8th week
Fetus	End of 2nd month to birth	Rapid growth in length and weight; peak length velocity at ~18 weeks; growth slows from ~36 weeks

Although the velocity of growth in body length is greater at birth than at any later period, in fetal life the velocity is greater still. Its peak is reached at about the eighteenth week of postmenstrual age. Growth in fetal weight follows a similar pattern, except that the peak velocity is arrived at later, usually at the thirty-fourth postmenstrual week.

From about 36 weeks to birth at 40 weeks, the rate of growth slows down, perhaps because the space available in the uterus is becoming fully occupied. Birth weight and birth size reflect the maternal environment more than the child's genotype. Directly after birth, the growth rate increases again, particularly in genetically large children.

During the embryonic period, differentiation ('regionalization') occurs—the originally homogeneous fetus differentiates into regions such as the head and arms, and histogenesis (differentiation of cells into specialized tissues like muscle or nerve) takes place. The proportion of cells undergoing division decreases as the fetus gets older, and hardly any new muscle or nerve cells appear after the sixth month of fetal life (Jaiswal, 2013).

Check Your Progress 3

3) What are the main characteristics of Prenatal Growth?

Section 2.4

POSTNATAL GROWTH

Postnatal growth starts from the time of birth up to adulthood or the time of maturity. The postnatal period is commonly divided into the following age periods: Neonatal, Infancy, Childhood, Adolescence, Adulthood/Maturity, and Senility or old age.

2.4.1 Neonatal Stage

The neonatal stage begins with birth and ends when the infant is approximately two weeks old. It is the shortest of all developmental periods. It is a time when changes are very rapid. Many critical events occur: feeding patterns are established, bonding between parents and infant begins, the risk for infections is higher, and many birth or congenital defects are first noted.

2.4.2 Infancy Stage

Infancy lasts from two weeks to one year of life. It is the true foundation period of life, characterized by rapid change in bodily systems, psychological development and neuromuscular organization. During infancy, growth is extremely rapid—more than 50% of birth length and 200% of birth weight occur during the first year of life. From this period, the rate of growth decreases sharply. Birth length represents approximately 1/4 of the total stature (against 1/8 in the adult). Cranial capacity is about 1/2 of the adult (Jaiswal, 2013).

2.4.3 Childhood

Childhood ranges from the end of infancy up to the beginning of adolescence. Both heredity and environment play their respective roles in physical growth. The hereditary growth potential of a child may be influenced by various environmental factors including nutrition, state of health, socio-economic condition, and psychological well-being.

Phase	Age Range	Key Characteristics
Early Childhood	1–6 years	Eruption of deciduous (milk) teeth; rounded body form persists; head predominates over trunk; more growth in width than height
Middle Childhood	7–10 years	Permanent teeth erupt; rounded forms disappear; linear growth rapid; thorax becomes oval; mid-growth spurt at 7–8 years
Late Childhood	Prepubertal period	Growth rate decreases and becomes negligible; sexual morphological differentiation begins; adolescent growth spurt observed

2.4.4 Adolescence

Adolescence is the threshold of adulthood. It extends from twelve years to generally eighteen to twenty years—starting from the time of puberty and continuing up to the onset of adulthood. The acceleration of growth at adolescence causes many anatomical changes in almost all parts of the body. Sexual maturation takes place during this period.

This period may be divided into **prepuberal** and **puberal** phases. During the prepuberal phase (about two years), weight increase is retarded, height increases by lengthening of the legs, and the thorax becomes narrow. During the puberal period, sexual organs mature, body proportions change, and secondary sexual characters appear. In girls, the first menstruation marks the beginning of ovarian activity. The most obvious secondary sexual characters in girls are breast development and appearance of axillary and pubic hair. In boys, pubic hair, beard, and moustache appear, and the voice changes (Jaiswal, 2013).

2.4.5 Maturity or Adulthood

Adults are individuals who have completed their growth and are ready to assume their status in society. The cessation of growth of stature is viewed as a sign of maturity. Height ceases to grow when the long bones (e.g., tibia, fibula) lose their capacity to increase in length. Usually, males attain adult height at about 21 years and females at 18 years of age.

Adulthood is the longest period of the lifespan—a period of more or less stability. It is further divided into: **Early adulthood** (18–20 to 40 years), **Middle adulthood** (40 to 60 years), and **Late adulthood or Senility** (60 years to death).

2.4.6 Senescence

After the active phase of the lifespan, decline processes start resulting in old age or senescence. The aging pattern shows great individual variation. During this period, numerous molecular and cellular changes happen. The processes of growing old are scientifically studied in **Gerontology**.

Because of aging, tissues do not renew and cells show senile involution. The memory declines. Systolic blood pressure increases. The range of accommodation of the eye lens declines. There is reduction of density of long bones and vertebrae, leading to decrease in height and sitting height. Vital capacity and muscle tone decline (Jaiswal, 2013).

Check Your Progress 4

4) Explain in brief about Postnatal Growth.

Section 2.5

NORMAL GROWTH CURVES PATTERN

Growth is measurable, and therefore, **anthropometry** plays a very important role in the study of growth. The measurements at successive ages are plotted in a graph to obtain growth curves. Usually two types of growth curves are drawn:

Distance Curve	Velocity Curve
Shows the total amount of growth achieved at each age—cumulative height or weight over time. The curve is S-shaped (sigmoid).	Shows the rate of growth at each age—how fast the body is growing at any given point. Reveals spurts and decelerations clearly.

De Montbeillard measured the height of his son at regular intervals (every 6 months) from his birth in 1759 until he was 18 years old in 1777. Richard E. Scammon plotted these measurements and published his findings in 1927 in the American Journal of Physical Anthropology.

*The human growth curve is characterized by **two sigmoid curves**—one in early life (the infant phase) and the other in the adolescent phase. They are separated by a period of relatively slow growth rate during childhood. The growth curve thus has **four distinct phases**: rapid growth in infancy, steady growth in childhood, rapid growth during adolescence, and slow growth as the individual approaches adulthood. A rapid growth during childhood known as the **mid-growth spurt** is also observed but is not a universal phenomenon.*

Check Your Progress 5

5) Write short notes on Pattern of Normal Growth Curves.

Section 2.6

NORMAL GROWTH VARIATIONS

Variations in normal growth represent alternative pathways that enable individuals to achieve their growth potential. The S-shaped (sigmoid) curve represents four phases of growth (Malina et al., 2004): rapid growth in infancy and early childhood; steady growth during middle childhood; rapid growth during adolescence; and slow and eventual cessation of growth after adolescence continuing into the third decade of life.

2.6.1 Canalization

Human growth and development is a complex phenomenon regulated by both genetic and environmental factors. At conception, an individual obtains a genetic blueprint which determines the potential for achieving adult size and shape. In an unconstrained environment, an individual would

display a pattern of growth that is more or less parallel to the genetically predetermined trajectory. This phenomenon was described as "**canalization**" or "**homeorrhexis**" by British geneticist **C.H. Waddington** (1957). However, none of us has been raised in a totally unconstrained environment.

2.6.2 Catch-Down Growth

During prenatal and postnatal life, we are exposed to an array of adverse environmental factors (such as diseases, malnutrition, etc.) which influence the growth velocity. It either slows down (catch-down growth) or in more severe cases would cease (Cameron, 2012). The amount of deviation from the predetermined pathway depends on the frequency, duration, and severity of exposure. The extent of influence is also determined by the age and gender of the individual (Tanner, 1978).

2.6.3 Catch-Up Growth

Even after being pushed off the trajectories, there exists an ability to stabilize and return to a predetermined growth curve when conditions improve. During such recovery, initial growth velocity is above that normally expected for children of that age. This rapid increase in growth velocity following a short-term period of starvation or illness was termed "**catch-up growth**" by **Prader et al.** (1963).

Catch-up growth may completely restore the situation to normal, or it may be insufficient to do so. If incomplete, the individual does not achieve the same adult height that would have been achieved without growth impairment. This depends on the extent to which conditions are optimized during recovery (McCarthy et al., 2013).

Check Your Progress 6

6) What are the different types of variation from normal growth?

Section 2.7

GROWTH CURVES AND ETHNIC AND GENDER DIFFERENCES

Growth and development is the key indicator of biological maturity. There are two basic determinants of development: heredity and environment. A complicated mixture of these influences probably acts throughout the whole period of growth. Population differences—ethnic and gender differences—in the rate and timing of growth reflect differences in nutrition, environment, socio-economic status, and genetics.

Rapid economic and social transition in India has influenced growth patterns of children, especially in urban areas. Recent Indian studies suggest a trend towards increase in height (especially in boys) and an increase in obesity in both genders (Khadilkar et al., 2007; Ranjani et al., 2016). The Government of India accepted the WHO 2006 growth standards for monitoring growth of children under 5 since 2010. However, many studies suggest that these standards remain aspirational and many children get diagnosed as undernourished when they may be growing appropriately for the given population.

Section 2.8

SECULAR TREND

Secular trend is the trend observed over a span of several generations. It refers to the progressive pattern of change (over successive generations) in the age at which physical and behavioral milestones are achieved. Over the last 100 years, children of industrialized countries have acquired larger body size and taller stature more quickly than before; they reached sexual maturity more rapidly than their counterparts in other underprivileged countries. This is referred to as **"The Secular Trend."**

Studies by Roche (1979), Loesch et al. (2000), and others on countries such as Germany, England, Poland, Norway, Sweden, and North America show that children of these countries have a marked tendency to become progressively taller and heavier at all ages. The secular trend in age at menarche has also been noted—among girls in those countries, the age at menarche was gradually decreasing from 1880 to 1980.

What are the reasons behind the secular trend? There is no definite answer. Better nutrition, improved health facilities, desirable environmental conditions, and marriage distance in terms of geography as well as mating outside the groups may be contributing factors (Jaiswal, 2013).

Check Your Progress 7

7) Define the concept of Secular Trend.

Section 2.9

SUMMARY

1. Primates generally have a slow reproductive turnover and drawn-out life histories.
 2. Growth refers to quantitative changes (size, weight, length), while development refers to qualitative changes (structure, organization, function).
 3. The two major periods of growth are prenatal and postnatal.
 4. Prenatal development comprises three stages: zygote, embryo, and fetus.
 5. Postnatal growth is divided into: neonatal, infancy, childhood, adolescence, adulthood, and senescence.
 6. Growth curves include distance curves (cumulative size) and velocity curves (rate of growth).
 7. The human growth curve has a characteristic double sigmoid shape with infant and adolescent growth spurts.
 8. Canalization describes genetically predetermined growth trajectories; catch-up and catch-down growth are deviations from these trajectories.
 9. Ethnic and gender differences in growth reflect variations in nutrition, environment, socio-economic status, and genetics.
 10. The secular trend refers to progressive changes across generations in the achievement of physical and behavioral milestones.
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Section 2.10

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Section 2.11

ANSWERS TO CHECK YOUR PROGRESS

- 1) Growth means increase and enlargement of the body or some parts of the body (quantitative changes that can be observed and measured). Development refers to qualitative changes—changes in kind, structure, or organization. Growth can be one of the components of development, or development in its quantitative aspect can be termed as growth. (Refer to Section 2.1)
- 2) The two major periods of growth are **prenatal** and **postnatal**. (Refer to Section 2.2)
- 3) At conception, the human ovum is about 0.1 mm in diameter. During the ovum stage it resembles a homogeneous mass. During the embryonic stage, differentiation begins to form various body regions (morphogenesis). By the eighth week the embryo appears childlike. During the fetus stage, growth in length and weight is rapid. Length peak velocity occurs at about 18 weeks. From 36 weeks, fetal growth slows due to uterine space constraints. (Refer to Section 2.3)
- 4) Postnatal growth spans from birth to maturity. It is divided into: Neonatal (first 2 weeks), Infancy (2 weeks to 1 year), Childhood (1 year to adolescence), Adolescence (puberty to 18–20 years), Adulthood/Maturity (18–20 to 60 years), and Senility (after 60 years). (Refer to Section 2.4)
- 5) Growth curves include distance curves (showing cumulative size at each age) and velocity curves (showing rate of growth at each age). Anthropometry is key to growth studies. (Refer to Section 2.5)
- 6) Normal growth variations include: canalization (growth along genetically predetermined trajectories), catch-down growth (slowing of growth due to adverse environmental factors), and catch-up growth (rapid recovery in growth velocity when conditions improve). (Refer to Section 2.6)
- 7) Secular trend is the progressive pattern of change (over successive generations) in the age at which physical and behavioral milestones are achieved. Over the last 100 years, children in industrialized countries have become taller, heavier, and reached maturity earlier. (Refer to Section 2.8)